Available online at www.sciencedirect.com
SCIENCE DIRECT®
Journal of Asian Earth Sciences 24 (2005) 481–493
Journal of Asian Earth Sciences
www.elsevier.com/locate/jaes

New and revised set of Cretaceous paleomagnetic poles from Hong Kong: implications for the development of southeast China

Yongxiang Li[a,*], Jason R. Ali[a], Lung S. Chan[a], Cho Min Lee[a,b]

[a]*Department of Earth Sciences, The University of Hong Kong, Pokfulam Road, Hong Kong, China*
[b]*Department of Civil and Structural Engineering, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong, China*

Received 15 April 2003; revised 10 December 2003; accepted 12 January 2004

Abstract

A paleomagnetic study of Upper Cretaceous (~80 Ma) dikes from Hong Kong was conducted with the aim of better understanding the Late Cretaceous–Cenozoic kinematic history of southeast China. Twelve out of seventeen sites collected from four dikes revealed primary TRMs and yield a Late Cretaceous paleopole at 69.3°N/211.2°E, where $A_{95} = 8.9°$ and $K = 107$. A synthesis of South China Block Cretaceous poles indicates that Coastal Southeast China, which is bounded by the Lianhuashan Fault Zone to the northwest and the Changle-Nanao Fault Zone to the southeast, has moved relative to inland Central South China. The discordant Cretaceous paleopoles from the South China Block suggest a moderate net clockwise rotation of the Coastal Southeast China relative to Central South China since the Late Cretaceous. This clockwise rotation is probably the result of micro-blocks along the South China margin experiencing vertical-axis rotations in response to the penetrative dextral shear along the SE China margin during oblique subduction of the Pacific Plate beneath South China. An alternative interpretation involves regional extension that displaced a raft-block southeastwards of the margin, generating apparent paleomagnetic records with no real block rotations.
© 2004 Elsevier Ltd. All rights reserved.

Keywords: Paleomagnetism; Vertical-axis rotation; Extension; Southeast China; Hong Kong

1. Introduction

The South China Block (SCB), comprising the Yangtze Craton in the north and Cathysia block in the south, is situated at the southeastern tip of the Eurasia Plate (Fig. 1). South China began amalgamating with the North China Block in the late Permian, and had taken-up its basic current position within Eurasia by the middle Cretaceous (Enkin et al., 1992; Metcalfe, 1996; Zhao et al., 1996). During the Jurassic and Early Cretaceous, an Andean-type convergent margin existed in South China and extended a considerable distance from Indochina to northeast Asia (Li, 1998). In southeast China, vast amounts of intrusive and extrusive rocks were generated (Jahn et al., 1990; Charvet et al., 1994). In the middle Cretaceous, the tectonic regime switched to one in which local and regional extension dominated with grabens, pull-apart basins, and basaltic magmatism (Gilder et al., 1991; Lapierre et al., 1997; Li and McCulloch, 1998; Zhou and Li, 2000). The Paleogene indentation of India into Eurasia resulted in western South China being deformed (Burchfiel et al., 1995) and affected by the extrusion of the Indochina block (Tapponnier et al., 1982, 1986). Paleomagnetic data from the sub-region have been used to quantitatively test the proposed tectonic models (Funahara et al., 1992; Otofuji et al., 1998; Sato et al., 1999; Yang et al., 2001).

Although it is assumed that the South China Block has largely behaved as a coherent block since ~100 Ma (Tapponnier et al., 1982; Liu and Morinaga, 1999), this idea has never been rigorously tested. In fact Enkin et al. (1992) and Gilder et al. (1993) have drawn attention to the streaked distribution of the Cretaceous paleopoles from the South China Block, which they argue could represent within-plate deformation-rotations.

Compared to the southwestern part of South China, the paleomagnetic database for the other sites within the block, particularly the 'Pacific' eastern margin, is somewhat limited. This sub-region is dominated by Yanshanian

* Corresponding author. Address. Department of Earth and Environmental Sciences, Lehigh University, Bethlehem, PA, 18015-3188, USA. Tel.: +1-610-758-3213; fax: +1-610-758-3677.
E-mail address: yol5@lehigh.edu (Y. Li).

1367-9120/$ - see front matter © 2004 Elsevier Ltd. All rights reserved.
doi:10.1016/j.jseaes.2004.01.004

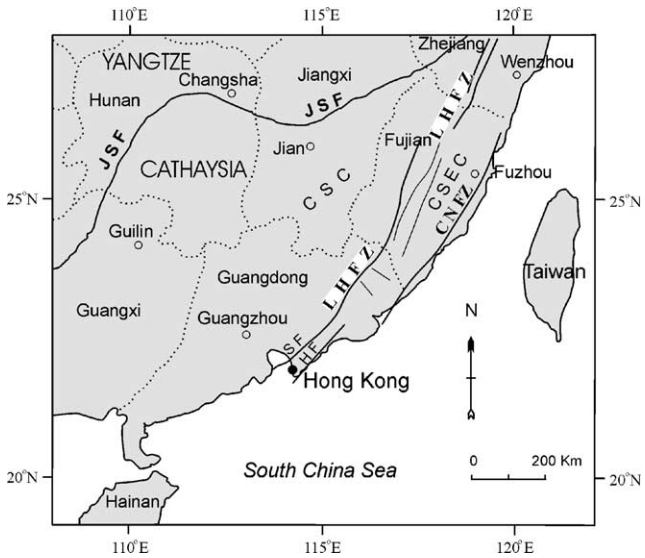


Fig. 1. Regional tectonic map of SE China. JSF, Jiangshan-Shaoxin Fault; LHFZ, Lianhuashan Fault Zone; CNFZ, Changle-Nanao Fault Zone; SF, Shenzhen Fault; HF, Haifeng Fault; CSEC, Coastal SE China; CSC, Central South China (after Davis et al., 1997).

(from ~180 to ~70 Ma) (Weng, 1927) igneous rocks. The Cretaceous igneous rocks in the area are exposed between two NE-striking, margin-parallel fault zones, the Lianhuashan Fault Zone (LHFZ) to the northwest and the Changle-Nanao Fault Zone to the SE (CNFZ) (Fig. 1). These two fault zones in effect define the Coastal SE China (CSEC) sliver block, which has/had a configuration not unlike a mirrored image of Sumatra (McCarthy et al., 1997) in eastern Indonesia. Thus regional stress could have resulted in movement of the CSEC relative to the main South China Block. Although several paleomagnetic studies have been done in this region (Chan, 1991; Gilder et al., 1993; Liu et al., 1999), few studies have systematically investigated the kinematics of the CSEC and none have attempted to understand the mechanisms driving block movement.

This study presents paleomagnetic results from Upper Cretaceous dikes in Hong Kong. Paleomagnetic data from granitic rocks (Chan, 1991) are reinterpreted following the systematic age-dating studies (Davis et al., 1997; Sewell and Campbell, 1997; Campbell and Sewell, 1998; Sewell et al., 1998), which have led to a revised stratigraphic scheme for the Hong Kong Special Administrative Region of China. The new results and the revised data are used to delineate the post Early Cretaceous block rotations and are compared with published data obtained from other parts of the sub-region.

2. Sampling and laboratory techniques

Samples for the present study were collected from Hong Kong, which sits at the southwestern end of the CSEC (Fig. 1). Hong Kong became the ideal area for studying the paleomagnetic-kinematics of SE China because the very detailed geological mapping and intensive dating work have been conducted in this region over the past several years. Together with these additional constraints, the paleomagnetic study of this region is expected to elucidate the kinematics of SE China. The hard-rock geology of the territory can essentially be split into three age-related packages: (i) deformed Devonian through Early Jurassic shelf sequence; (ii) Middle Jurassic to Early Cretaceous continental arc sequence; (iii) post-arc terrestrial/continental margin sequences spanning the Early Cretaceous–Paleocene (Sewell et al., 2000).

Paleomagnetic samples were drilled from Upper Cretaceous dikes at two localities: one is located southeast of Lamma Island and the other is next to the East Dam in High Island (Fig. 2a). On Lamma Island, three cross-cutting dikes were sampled: Lamma basaltic dike-1 (BD1); Lamma aplite dike (AD), and Lamma basaltic dike-2 (BD2) (Fig. 2b). Three sites were collected from each dike. One was collected from the edge of BD2 in granitic country rocks and another was sampled from granitic country rocks far away from the dikes in order to perform a baked contact test. The youngest basic dike sub-vertically cuts the South Lamma Granite. Since Lamma Island is dominated by massive granitic rocks, it was not possible to measure a 'bedding' attitude directly in the field. Instead, the orientation of the youngest basic dike was measured and it was assumed that the dike intruded the granitic country-rock vertically. Thus, the plane perpendicular to the youngest basic dike was considered to be the paleohorizontal and was used for tilt correction. All samples were collected from the cliff except Sites 5, 6, and 7. These three sites were sampled from what were considered along-strike equivalents of BD1 and AD on the platform, which is separated from the cliff by numerous boulders. At the High Island East Dam, five sites were sampled from different parts of a 1-m-wide mafic dike and one site from the columnar-jointed rhyolite country rocks it intrudes. The dike is tilted and its contact with the rhyolite country rocks is sharp. It is assumed that the columnar joints were initially orthogonal to the paleohorizontal. We measured the orientations of two sets of joints exposed on the cliff. These measurements allow us to find the pole to paleohorizontal for tilt correction (Fig. 3). Six to eight cores were drilled from each site and oriented with a Brunton compass mounted on a Pomeroy Orientation table.

All samples were processed and measured at the paleomagnetic laboratory of the University of Hong Kong. Samples were subject to either detailed progressive thermal or alternating field (AF) demagnetization in 12 to 15 steps up to 600 °C or 100 mT. A Magnetic Measurements MMTD1 thermal demagnetizer was used for thermal demagnetization and a Molspin AF tumbling demagnetizer was utilized for AF demagnetization. A Molspin spinner magnetometer was used to measure remanence. Characteristic remanent magnetization directions (ChRM) were calculated using the principal component analysis techniques of Kirschvink (1980) and the site-mean results were analyzed using Fisher statistics (Fisher, 1953). NRM/IRM AF demagnetization

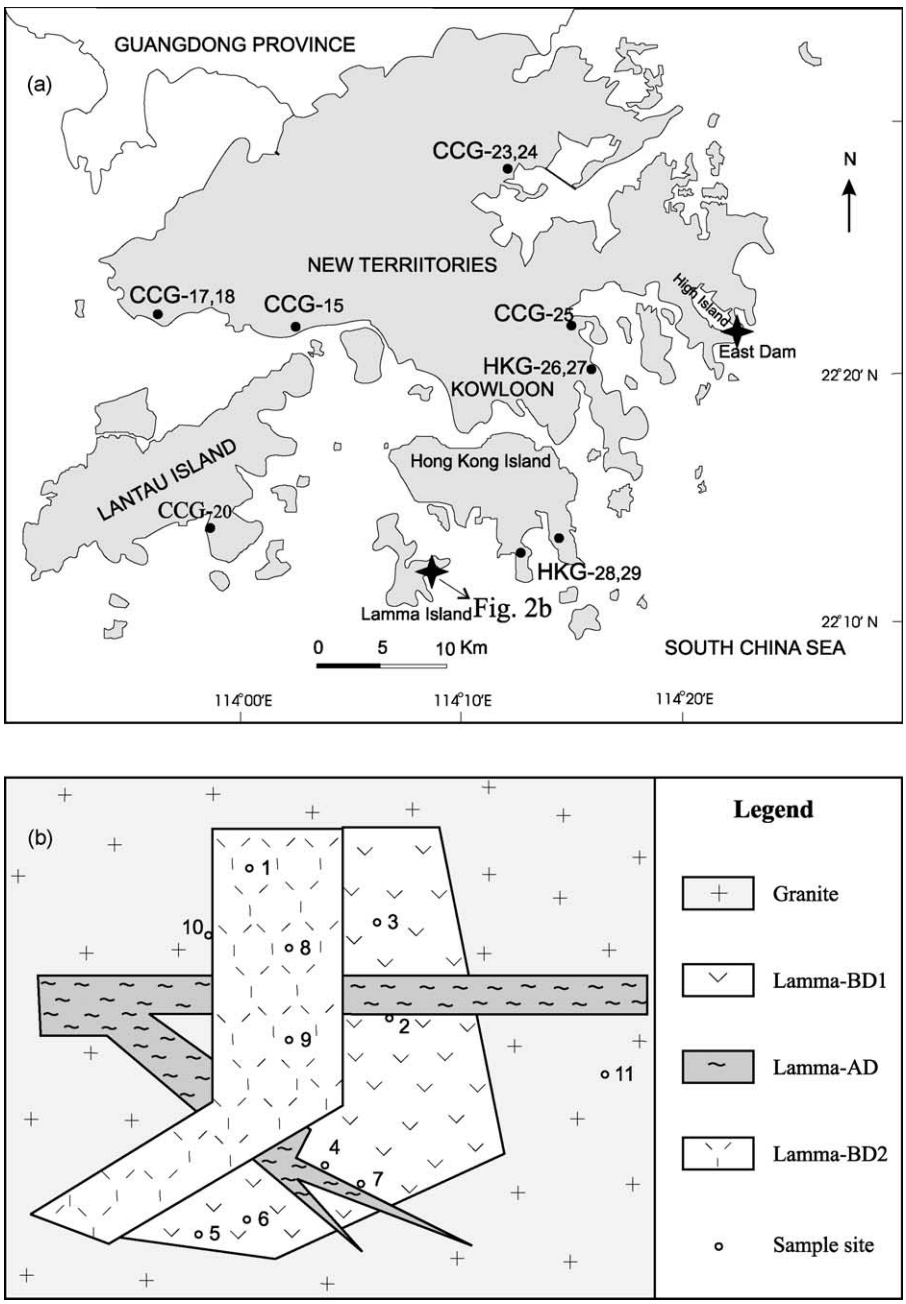


Fig. 2. (a) Simplified map of Hong Kong showing the sampling localities indicated as stars. The dots stand for sampling sites that yield paleomagnetic data rated at 'A' and 'B' quality by Chan (1991) (see Table 3 for the paleomagnetic data). (b) Schematic diagram shows the cross-cutting relationships of three dikes on Lamma Island.

tests of Fuller et al. (1988) and Cisowski et al. (1990) were employed to deduce whether the ChRMs were primary or not. A fold test of Mcfadden and Jones (1981) was applied to further constrain the age of remanence.

3. Results

Table 1 summarizes the paleomagnetic results from this study. Seven of the eleven Lamma Island sites yielded stable ChRMs, often after a viscous component was removed at 10–20 mT. Reversed components were identified from all BD2 samples and are approximately antipodal with the normal polarity component isolated from other sites (Fig. 4a and b). Magnetite was regarded as the principal remanence carrier, as indicated by the 580 °C unblocking temperature (Fig. 4b). Site 10 was taken from the baked contact of BD2 in the granitic country rocks and showed similar directions to those BD2 samples. Therefore, Site 10 is considered to carry the same remanence as BD2 and was included for calculating the mean ChRM for BD2. Site 11 came from the granitic country rocks and

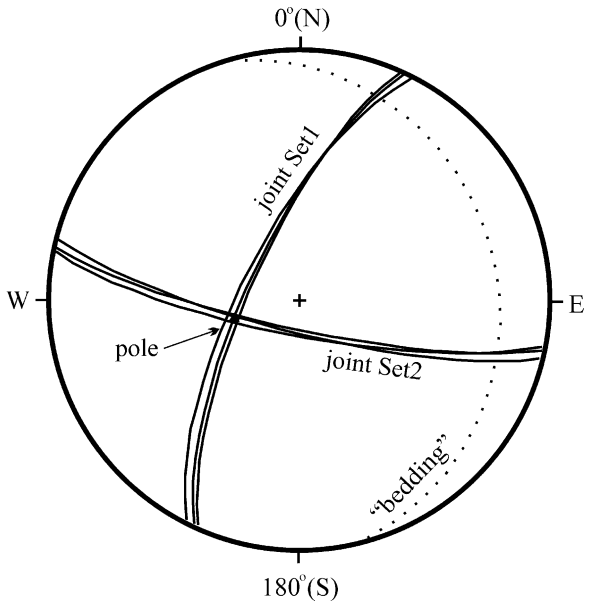


Fig. 3. The stereographic projection of the orientations of two sets of joints in rhyolite country rocks at East Dam. The solid circles represent the orientations of these two sets of joints and the intersection of these solid circles defines the pole to 'bedding'. The dashed circle represents the bedding (344°/21°E), which is considered to be the paleohorizontal and was used for tilt correction.

yielded erratic directions, which is probably due to their coarse grains. Sites 5, 6, and 7 revealed large within-site dispersion of directions, with confidence circle greater than 30°, and were excluded from further analyses. Overall, the demagnetization behavior of Lamma Island dikes indicates that BD2 has higher coercivities than BD1 and AD. Also, no overprints of BD1 and AD by the BD2 magnetization were observed.

Stable magnetizations with normal polarity were isolated from all five sites taken from the dike in East Dam after the viscous components were removed, typically at ~5 mT (Fig. 4c). Magnetite was considered to be the major remanence carrier, as indicated by the relatively low magnetic coercivities (e.g. Fig. 4d). The site sampled in the rhyolite country-rocks did not yield stable directions.

The mean ChRM directions for four cooling units (sometimes several sites/dike) are well grouped in the geographic coordinates and all depart from the time-averaged field at the sampling sites (Fig. 5a). The clustering of mean directions was improved after tilt correction (Fig. 5b). Owing to the weak magnetization and erratic demagnetization directions of the country rocks, it was not possible to apply a baked contact test to constrain the age of remanence. However, IRM AF demagnetization

Table 1
Paleomagnetic results from the Upper Cretaceous dikes from Lamma island and East dam in high island

UNIT	Site no.	Site location		'Bedding'	*n*/*N*	In situ mean		Tilt-corrected				Paleopole		
		Lat. d (°N)	Long. (°E)	Strike / Dip		Dec	Inc	Dec	Inc	Dispersion		P. Lat.	P. Long.	d*p*/d*m*
										k	α_{95}/Sep.			
Lamma Basaltic Dike-2 (Lamma-BD2)	Lamma1	22.2	114.14	245/9	12/12	214.0	−49.0	206.2	−43.8	119.9	4.0	65.8	190.8	
	Lamma8	22.2	114.14	245/9	12/12	220.0	−42.0	213.4	−37.7	694.0	1.6	59.0	199.7	
	Lamma9	22.2	114.14	245/9	5/3	222.3	−47.4	214.1	−43.3	432.6	5.9	58.7	191.7	
	Lamma10	22.2	114.14	245/9	7/4	217.7	−45.1	210.5	−40.5	87.2	13.3	61.9	196.3	
	Mean				4	218.6	−45.9	211.1	−41.4	435.7	4.4	61.4	196.3	3.3/5.4
Lamma − BD1	Lamma3	22.2	114.14	245/9	8/8	22.0	34.2	18.2	27.8	26.5	11.0	71.2	224.3	6.6/12.0
Aplite dike (Lamma-AD)	Lamma2	22.2	114.14	245/9	7/6	14.4	40.5	10.3	33.3	44.3	10.2	79.5	225.8	
	Lamma4	22.2	114.14	245/9	9/6	22.7	41.0	17.7	34.6	45.2	10.1	73.1	211.7	
	Mean				2	18.5	40.8	14.0	34.0		/6.3	76.4	216.8	
Dike at East Dam (ED)	ED1	22.37	114.38	344/21	9/9	17.5	43.2	29.6	29.7	182.7	3.8	61.4	212.1	
	ED2	22.37	114.38	344/21	7/6	14.9	46.5	29.0	33.4	145.6	5.6	62.5	207.7	
	ED3	22.37	114.38	344/21	9/9	8.6	45.2	23.5	30.6	53.1	7.1	67.1	215.1	
	ED4	22.37	114.38	344/21	9/9	6.9	41.1	20.3	30.6	244.6	3.3	70.0	217.9	
	ED5	22.37	114.38	344/21	7/7	5.4	35.3	16.6	25.6	168.8	4.7	71.9	231.0	
	Mean				5	10.5	42.4	23.7	30.1	208.3	5.3	66.8	215.1	3.3/5.9
Mean of above 12 sites					12	21.8	43.2	23.8	34.4			67.3	207.9	$A_{95} = 4.4$
Mean of above 4 units					4						$k = 107$	69.3	211.2	$A_{95} = 8.9$

Note: Site No. = site number; Lat.(°N) = (northern) latitude of sample site; Long..(°E) = (eastern) longitude of sample site; Strike/Dip = orientation of bedding following the right-hand rule; *n*/*N* =number of demagnetized samples/number of samples used to calculate the mean ChRMs; Dec = Declination; Inc = Inclination; *k* =Fisher (1953) precision parameter; α_{95}/Sep. = radius of the 95% confidence circle/Separated angle for two-site mean; P Lat. = (northern) latitude of pole; P. Long = longitude of pole; d*p*/d*m*, semi-axes of the elliptical error around the pole at the 95% probability, d*p* in the colatitude direction and d*m* perpendicular to it; All angles are in degrees.

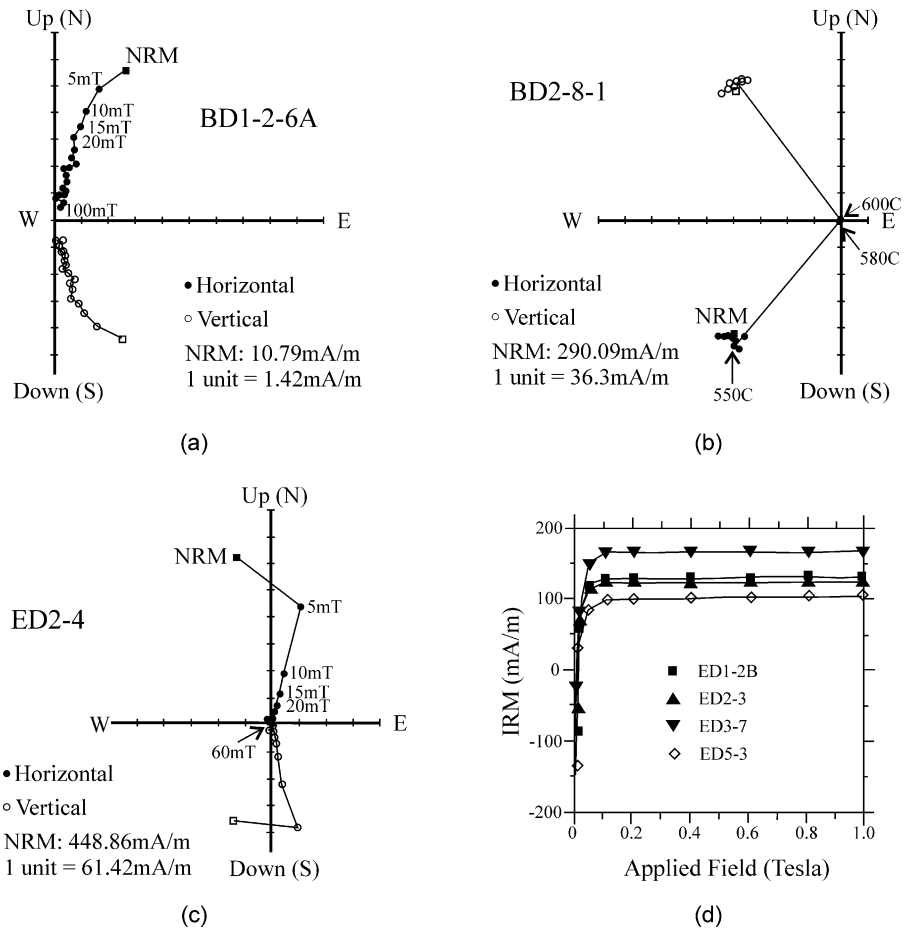


Fig. 4. Representative af and thermal demagnetization plots (Zijderveld, 1967) from Lamma Island (a, b) and East Dam dikes (c). The back-field IRM acquisition curves (d) of East Dam dikes indicate that remanence was primarily carried by magnetite.

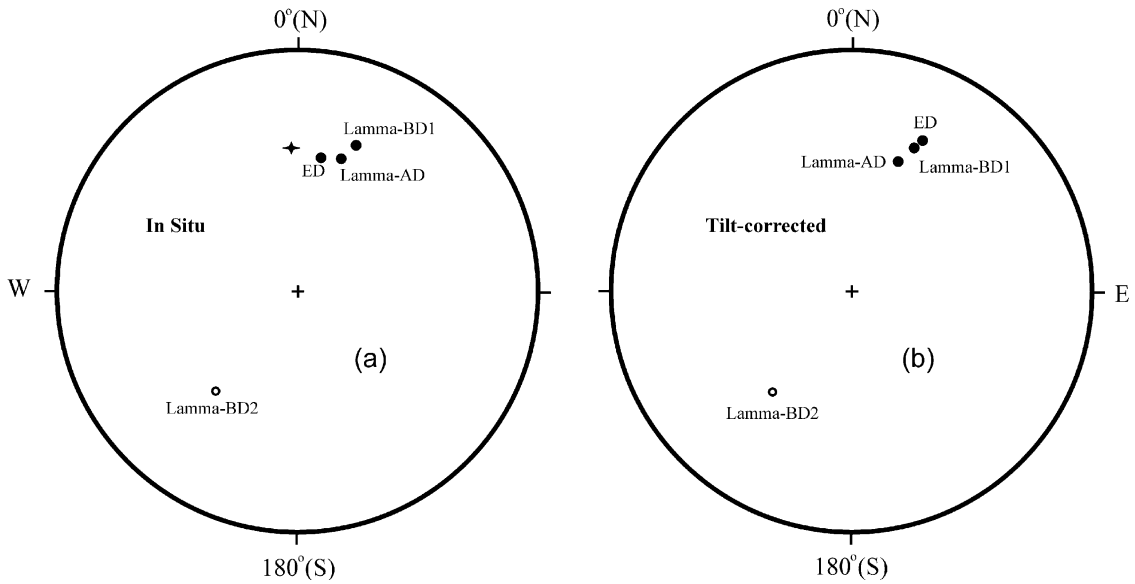


Fig. 5. The mean directions for four dikes in in situ coordinates (a) and in stratigraphic coordinates (b). Solid circles represent the downward direction and open circles indicate upward directions. (a) The mean directions depart from the present day field of the sampling site (star). (b) Tilt-correction improved slightly the clustering of mean directions. An antipodal of the reversed polarity was used to calculate the precision parameter.

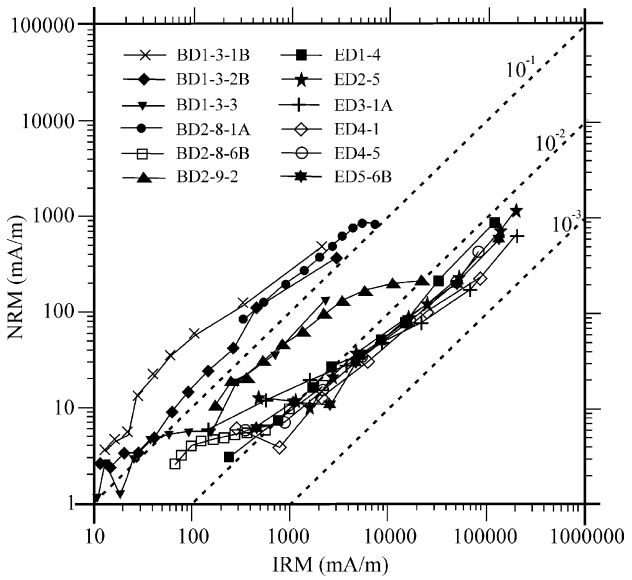


Fig. 6. Representative NRM/IRM-AF demagnetization data of Lamma Island and East Dam dikes suggest that remanences are primary TRMs (Fuller et al., 1988; Cisowski et al., 1990).

was performed on representative samples and the NRM-IRM AF demagnetization plots were then used to estimate if the ChRMs were primary (Fig. 6). The principal feature of these experiments is that the NRM/IRM ratios at each demagnetization step are either greater than or close to 10^{-2} and the NRM-IRM AF demagnetization paths suggest that dike magnetizations are primary TRMs (Fuller et al., 1988; Cisowski et al., 1990). Because the grouping of the directions is only slightly improved after tilt correction (Fig. 5), we further performed a fold test (Mcfadden and Jones, 1981) by considering the Lamma Island and East Dam sites as two limbs of a broad-scale fold. The calculation indicates that a true mean is rejected in geographic coordinates and a true mean is accepted in stratigraphic coordinates at the 95% confidence level (Table 2), yielding a positive fold test. Note that this fold test is not very robust because of the small contrast in the beddings of two limbs of the fold. Nevertheless, the increase in the concentration of directions after tilt-correction supports the interpretation that the remanence is primary in origin.

Table 2
A fold test (Mcfadden and Jones, 1981) for the Upper Cretaceous dikes from Lamma Island and East Dam in High Island

Limb	Number of sites	in situ	Tilt-corrected
Limb a (Lamma Island)	7	$R_a = 6.9212$	$R_a = 6.9211$
Limb b (East Dam)	5	$R_b = 4.9789$	$R_b = 4.9808$
		$R = 11.81$	$R = 11.877$
For $N = 12$ and $p = 0.05$ [Ra + Rb − R*R/ (Ra + Rb)]/2(N − Ra − Rb)		$= 0.89919 >$ 0.3493 True mean is rejected	$= 0.2539 <$ 0.3493 True mean is accepted

Note: the declinations and inclinations of individual sites are presented in Table 1.

4. Interpretation

4.1. The timing of remanence acquisition

The ages of the mafic dikes are constrained by (i) the age of the country rocks and (ii) correlation with similar dikes elsewhere in Hong Kong where age data are available. The granite on Lamma Island yields an U–Pb age of about 142.7 Ma and the rhyolite at East Dam in High Island yields an U–Pb age of 140.9 Ma (Davis et al., 1997). Therefore, the sampled dikes must be Cretaceous or younger. Allen and Stephens (1971) reported a 60 Ma K–Ar age for a mafic dike in Kowloon, Hong Kong. The radiometric data from mafic dikes from coastal areas of Fujian and Guangdong provinces have revealed that the mafic intrusions were emplaced primarily between 110 and 84 Ma (Chen et al., 1999). Geochemical studies of mafic dikes of the adjacent Guangdong Province indicate four Cretaceous extension–intrusion episodes along the South China margin region, the last one occurring at ~90 Ma (Li and McCulloch, 1998). A recent geochronologic study of mafic dikes in Hong Kong, including the dike sampled from the East Dam at High Island, yielded Ar–Ar dates ranging from 108.3 ± 1.6 to 75.1 ± 0.8 Ma (Sewell et al., 2000). The age of remanence can be further constrained by the magnetic polarity data as the geomagnetic field remained in a normal polarity configuration from 118–84 Ma. Thus the youngest mafic dike on Lamma Island must have acquired its remanence sometime after 84 Ma.

The analysis of the above age data led us to suggest that the sampled dikes were emplaced between 110 and 80 Ma, and more likely between 90 and 80 Ma, a conclusion which is supported by unpublished data held by the Hong Kong Geological Survey (R. J. Sewell, pers comm., 2002). Thus, the sampled dikes are believed to carry Late Cretaceous magnetizations.

4.2. Possible pre-cretaceous counter-clockwise rotation

In the earlier study by Chan (1991), the Cheung Chau (CCG) and Hong Kong Granites (HKG) were thought to be of Late Cretaceous age. However, the past decade has seen a substantial increase in the amount of high-precision radiometric age data for Hong Kong, and within this context, Chan's earlier assumption needs to be reassessed. U/Pb dating and geochemical fingerprinting of granitic and volcanic rocks in Hong Kong suggest four main phases of magmatism took place between 165 and 140 Ma (Sewell

Table 3
The paleomagnetic data of Chan (1991) with new dates and associated suites based on the revised stratigraphy of Hong Kong

Site No.	Unit	U/Pb Age	Suites	Inc	Dec	α_{95}	P. Lat.	P. Long.	Rate
26	HKG	~140	Lion Rock Suite	61.9	22.1	39.8	62.9	151.1	B
27	HKG	~140	Lion Rock Suite	34.2	10.6	27.4	79.2	225.2	B
28	HKG	~140	Lion Rock Suite	50.0	55.5	25.9	40.4	182.5	B
29	HKG	~140	Lion Rock Suite	42.5	7.2	17.4	83.3	188.6	A
20	CCG	~143.7	Cheung Chau Suite	37.1	347.8	10.7	78.5	14.9	A
23	CCG	~146	Kwai Chau Suite	51.9	340.9				B
24	CCG	~146	Kwai Chau Suite	44.5	42.6	36.3	51.3	190.4	B
25	CCG	~146	Kwai Chau Suite	40.3	10.4	9.3	80.5	202.0	A
25E	CCG	~146	Kwai Chau Suite	52.6	5.9	22.6	78.6	140.0	B
15	CCG	~160	Lamma Suite	47.0	3.2	18.2	84.0	142.4	A
17	CCG	~160	Lamma Suite	43.5	2.3	12.2	86.8	154.2	A
18	CCG	~160	Lamma Suite	49.3	22.2	21.2	68.9	179.4	B

Note: HKG, Hong Kong Granite; CCG, Cheung Chau Granite; Inc, Inclination; Dec, Declination; α_{95}, radius of the 95% confidence circle; P. Lat., (northern) latitude of pole; P. Long.., longitude of pole; All angles are in degrees.

and Campbell, 1997; Sewell et al., 1998). Thus the Late Cretaceous ages Chan (1991) ascribed to several granite bodies from which palaeomagnetic data were obtained should be corrected and accounted for in any regional plate model.

Table 3 shows the paleomagnetic data of twelve sites that were rated by Chan (1991) at 'A' and 'B' quality and were used for calculating a Late Cretaceous paleopole for Hong Kong. Besides the new dates, the new suites names are also presented, based onCampbell and Sewell (1998). Additionally, because all of Chan's 'B' class had α_{95} values $>20°$, these data have been excluded because they are considered unreliable (Van der Voo, 1990). Therefore only the 'A' quality data were used.

Chan (1991) 'A' quality data from the granite body and our new dike results are presented in Fig. 7. The obvious feature to emerge is that the mean directions of the two rock types are quite different. The granites have a mean direction centered on Dec = 2.0°, Inc = 42.4°, with $\alpha_{95} = 7.2°$, indicating little net block rotation with respect to stable Eurasia. In contrast, the Upper Cretaceous dikes have a direction centered on Dec = 21.5°, Inc = 33.5°, with $\alpha_{95} =$ 9.6°, suggesting ~10° clockwise rotation when compared to the predicted directions for stable Eurasia (Besse and Courtillot, 1991). This directional separation implies that, at least, locally complex block movements occurred between the Early and Late Cretaceous (counterclockwise) and Late Cretaceous and present (clockwise) because the net effect is that the granitic intrusions are orientated in a similar direction to when they were formed.

Although the available paleomagnetic data from Hong Kong requires a pre-Cretaceous counter-clockwise rotation, we caution against extrapolating the movement to the entire CSEC because of the paucity of Jurassic and younger poles from this part of the South China Block. For example, the only available Late Jurassic pole is from the northeast part of the CSEC (Enkin et al., 1992) and it is indistinguishable from the Cretaceous paleopoles (Fig. 8), which is not

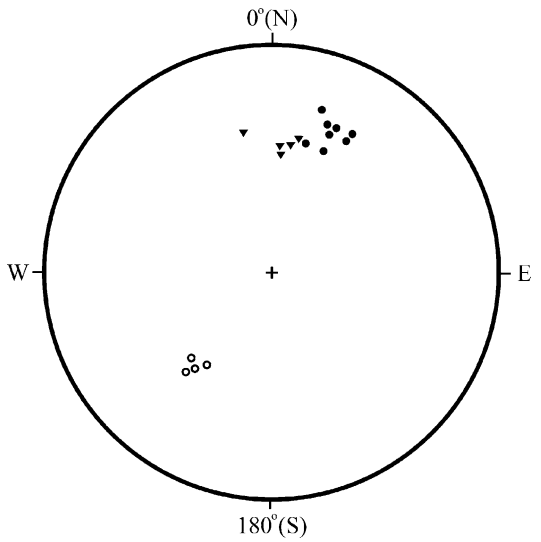


Fig. 7. Directional discrepancy of remanences of granitic rocks (triangles) and dikes (dots) indicates that relative block movements occurred between granitic intrusions and dike emplacement.

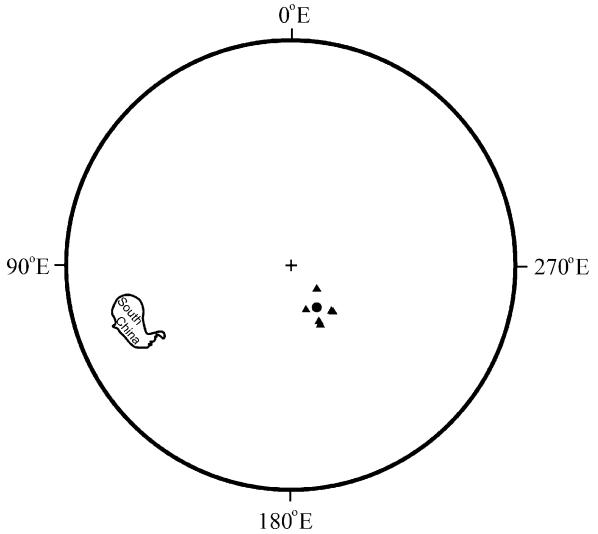


Fig. 8. The Late Jurassic paleopole (dots) from the northeast part of CSEC and the Cretaceous paleoples (triangles) of CSEC are indistinguishable.

Table 4
Cretaceous paleopoles from the CSEC and the CSC, separated by the Lianhuashan Fault Zone (LHFZ)

Age	Locality			VGP		A95 (dp/dm)	K	Reliability							Q	Sources	
	Lat.	Long.	$S[n]$	Lat.	Long.			1	2	3	4	5	6	7			
CSEC																	
$K2$	22.2	114.2	4	69.3	211.2	8.9	107.0		+	+	+	+	+	+	6	This study	
$K2$	26.0	117.3	[20]	65.1	207.2	5.0		+	+	+	+	+		+	6	Gilder et al., 1993	
$K2$	25.0	116.4		67.9	186.2	9.2		+	+	+	+	+	+	+	7	Hu, et al., 1990	In: Ma et al., 1993
$K2$	28.4	119.9	5	71.4	204.7	3.1/4.8	433.0	+	+		+	+		+	5	Liu et al, 1999*	
Mean $Q >= 4$			$n = 4$	68.8	202.4	5.6	273.5										
$K1$	28.4	119.9	6	66.8	206.4	5.6	105.5	+	+		+	+		+	5	Liu et al., 1999*	
$K1$	26.0	117.3	5	66.9	221.4	5.4		+	+	+	+	+		+	6	Zhai et al., 1992	
$K1$	29.6	120.5	7	77.1	227.6	5.5	120.0	+	+	+		+	+	+	6	Lin et al., 1984	In: Cheng et al., 1996
Mean $Q >= 4$			$n = 3$	70.5	216.9	10.6	137.1										
CSC																	
$K2$	26.6	102.4	3	78.9	186.6	4.3/6.7		+	+	+		+		+	5	Zhu et al., 1988	
$K2$	26.5	102.4	18	81.5	220.9	7.1		+	+	+	+	+	+	+	7	Huang and Opdyke, 1992	
$K2$	32.0	119.0	10	76.3	172.6	10.3		+	+	+	+	+		+	6	Kent et al., 1986	In: Ma et al., 1993
$K2$	22.2	108.7	9	79.4	7.1	10.0		+	+	+		+		+	5	Gilder et al., 1993	
$K2$	27.0	115.0		79.2	158.6	1.0		+	+			+	+	+	5	Gu and Gu, 1990	In: Ma et al., 1993
$K2$	23.0	115.0	6	66.0	221.5	3.0		+				+	+		3	Hsü,1987	In: Ma et al., 1993
$K2$	19.6	109.5		80.6	175.1			+					+		2	Yu et al., 1990	In: Ma et al., 1993
$K2$	27.9	102.3	8	80.5	187.3			No info available								Zhu	In: Zhai and Zhou, 1989
$K2$	25.7	101.2		74.2	188.0			No info available								Liang, 1990	In: Cheng et al., 1996
$K2$	29.9	103.1	7	83.0	188.7	8.4		+	+	+					3	Zhuang et al., 1988	
$K2$	29.9	103.1	8	80.4	208.2	7.3		+	+						2	Zhuang et al., 1988	
$K2$	29.3	103.8	[14]	53.2	179.1			+		+		+			3	Zhou, Z.X. 1988	In: Cheng et al., 1996
$K2$	30.8	118.2	10	83.8	200.3	14.6		+	+	+	+	+	+	+	7	Gilder et al., 1999	
$K2$	30.0	102.9		72.8	241.1	6.6		+	+	+			+	+	5	Enkin et al., 1991	
Mean $Q >= 4$			$n = 7$	83.1	200.8	7.6	64.0										
CSC																	
$K1$	27.9	102.3	2	77.4	196.2	11.5/18.3		+	+	+				+	4	Zhu et al., 1988	
$K1$	26.8	102.5	7	69.0	204.6	4.3		+	+	+		+		+	5	Huang and Opdyke, 1992	
$K1$	27.9	102.3	[11]	86.3	166.3	4.1/6.3		No info available								Zhu et al.	In: Zhai and Zhou, 1989
$K1$	30.0	102.9	23	74.5	229.0	2.7/4.7		+	+	+	+		+	+	6	Enkin et al., 1991	
$K1$	25.7	101.2		78.8	194.8	3.7		No info available								Liang 1990	In: Cheng et al., 1996
$K1$	28.1	114.5		65.0	176.0	9.3/12.9		No info available								Wu et al., 1988	In: Enkin et al., 1992
$K1$	N.GD.	[348]		82.6	244.6	0.7/1.1		No info available								Xing	In: Zhai and Zhou, 1989
$K1$	30.1	103.0	8	78.6	201.0	15.3		+	+	+				+	4	Zhuang et al., 1988	In: Yang et al., 2001
$K1$	22.7	108.7	8	86.5	26.4	6.8		+	+	+	+	+		+	6	Gilder et al., 1993	
$K1$	19.6	109.5		80.0	246.4			No info available								Yu, 1990	In: Zhao et al., 1996
Mean $Q >= 4$			$n = 5$	78.9	208.5	9.0	72.5										

Note: symbols and abbreviations as Table 1. CSEC, Coast SE China; CSC, Central South China; N.GD: northern Guangdong; K1, Early Cretaceous; K2, Late Cretaceous; $S(n)$, sites (samples); A95, radius of the 95% confidence circle; dp/dm, semi-axes of the elliptical error around the pole at the 95% probability. dp, in the colatitudinal direction and dm perpendicular to it; All angles are in degrees. Data reliability criteria: 1, age of rocks; 2, statistic precision, $k > 10$, $\alpha_{95} < 16.0^{\circ}$; 3, detailed demagnetization and isolation of magnetic components; 4, positive field stability tests; 5, structural controls; 6, presence of reversals; 7, lack of remagnetization (Van der Voo, 1990). * Recalculated from original publication.

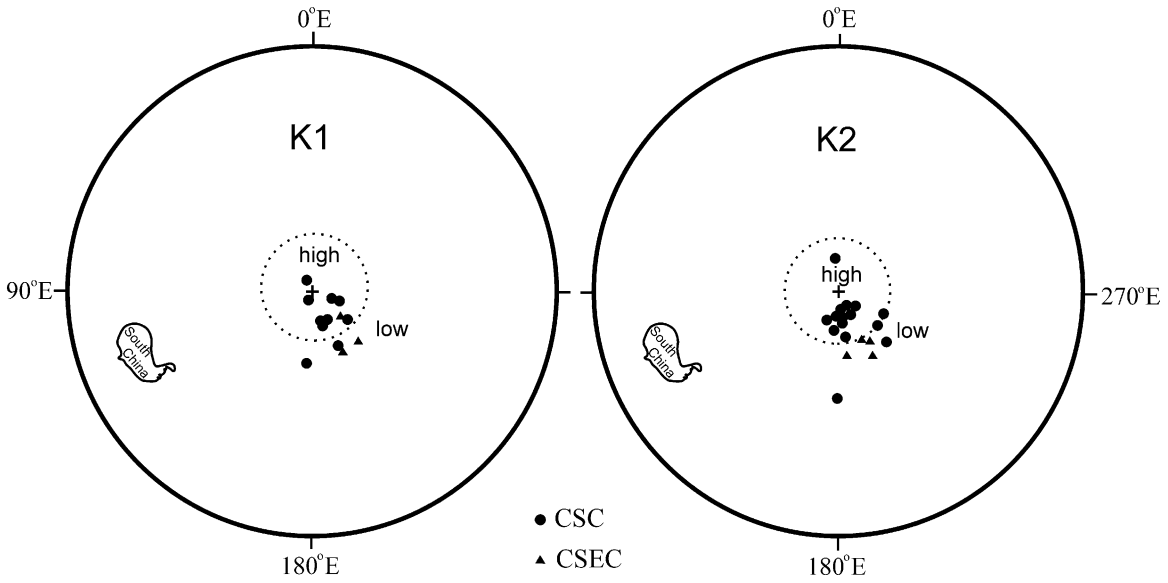


Fig. 9. The Early and Late Cretaceous paleopoles from the South China Block showing discordance between the CSC and the CSEC coeval paleopoles. The 70° latitudinal circle was marked on the plot separating the 'low'-latitudinal CSCE poles from the 'high'-latitudinal CSC poles. The separation of poles from the CSEC and the CSC may indicate relative block movements. K1, Early Cretaceous; K2, Late Cretaceous.

favorable to interpret the CCW rotation as a regional event. Thus, the paleomagnetically inferred pre-Cretaceous CCW rotation is preferentially considered to occur locally. The localized CCW rotation could be attributed to the sporadic sinistral strike-slip movement that is believed to have started as early as the Middle Jurassic (Sewell et al., 1998). More paleomagnetic data of Late Jurassic–Early Cretaceous age are required from other parts of CSEC to refine the interpretation of Cretaceous tectonic activity.

4.3. Clockwise rotation in southeastern China and possible mechanisms

Based on the IAGA palaeomagnetic pole database (http://dragon.ngu.no/Palmag/paleomag.htm), a series of Cretaceous paleopoles for South China were compiled (Table 4). Several additional paleomagnetic poles came from the Chinese literature. For readers' convenience, the original sources of all cited poles are also shown in Table 4. These poles are presented graphically in Fig. 9. The quality of these poles was carefully evaluated with the reliability criteria of Van der Voo (1990). While computing the mean pole for the blocks, poles with Q less than four were rejected. In order to examine whether the mobility of the CSEC was recorded in rocks from CSEC at the first order, paleopoles from the CSEC and the Central South China (CSC) are plotted with different symbols. The triangles denote paleopoles from the CSEC and the dots indicate paleopoles from the main part of the South China Block (Fig. 9). It appears that the majority (six out of seven) of the CSEC paleopoles lie in low latitudes; whereas, most poles from the CSC occur at high latitudes for both the Early and Late Cretaceous (Fig. 9), indicating a separation of coeval poles in the SCB. The apparent separation of the CSEC paleopoles from the coeval poles of the main body of the South China Block suggests that the plate is not a coherent block. Therefore, instead of calculating the mean pole for the SCB, the mean poles for the CSEC and the CSC were computed separately in order to estimate the amount of relative movements. The comparison of the averaged coeval poles for the CSEC and the CSC indicates that the CSEC may have rotated 11.6 ± 9.3° with respect to the stable CSC in a clockwise sense since the Late Cretaceous (Fig. 10).

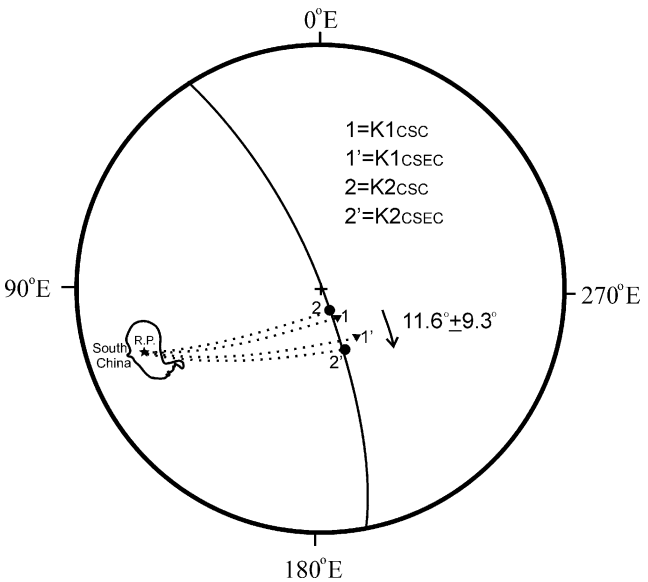


Fig. 10. Comparison of coeval poles from the CSEC and the CSC indicates that the CSEC may have undergone a 11.6 ± 9.3° clockwise rotation with respect to the CSC since the Late Cretaceous. The star represents the reference pole (R.P.) of the CSC (110°E, 25°N). The arc is part of the 65° colatitude small circle.

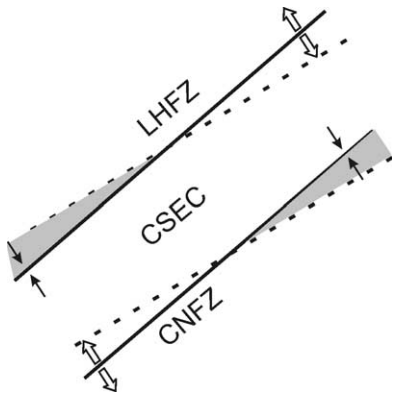


Fig. 11. Predicted compressional and extensional areas if the CSEC behaves as a single block. Shaded areas indicate the compressional region.

Another interesting feature about the CSEC poles is that the low-latitude standing CSEC paleopoles are more or less coincident (Fig. 9). The fact that the CSEC paleopoles cluster suggests that the CSEC sliver plate behaved as either a single coherent block or several microblocks that rotated in a similar sense since the Late Cretaceous. If the CSEC behaved as a single block, its clockwise rotation must result in extension in its NNE and SSE tips and a thrust in its NEE and SWW portions (Fig. 11). However, this prediction is not supported by structures in western Fujian, NNE tip of the CSEC, where thrusts are the dominant features (Chen, 1999). Therefore, it is likely that the CSEC sliver could have been broken into several microblocks located between the LHFZ and the CNFZ along the southeastern margin of South China. These microblocks may have undergone clockwise rotation about vertical axes since the Late Cretaceous. The breakup of the CSEC sliver and the vertical-axis clockwise rotation of these micro-blocks may be associated with dextral shear exerted along the southeastern margin due to the oblique subduction of the Pacific Plate underneath the SCB. Lu et al. 1993 and Wang and Lu 1997 carefully examined the shear fabrics of the CNFZ in eastern Fujian and postulated a micro-block collision model for the tectonic evolution of marginal South China. In this model, the Min(Fujian)tai(Taiwan) microplate (south flank of the CNFZ) collided with the Min(Fujian)zhe(zhejiang) microplate (≈ the CSEC sliver) at around 120-100 Ma. After this collision, the Mintai microplate, a passenger on the subducting Pacific Plate, has been driven to slip along the CNFZ in a dextral sense as the oblique subduction continued to generate shear along the coast. During the Cenozoic, the proto-Pacific Plate migration direction changed from NNW to NWW (Zhang, 1993), which led to a second phase of shearing in a dextral sense. The dextral shear along the SE China margin might have penetrated into the SCB, reached the LHFZ, and generated right-lateral movement along the LHFZ. The right-lateral strike-slip movement of the LHFZ in the Cenozoic has been well documented (Lin et al., 1984; Lin and Xie, 1996). Therefore, the penetrative dextral shear along the southeastern

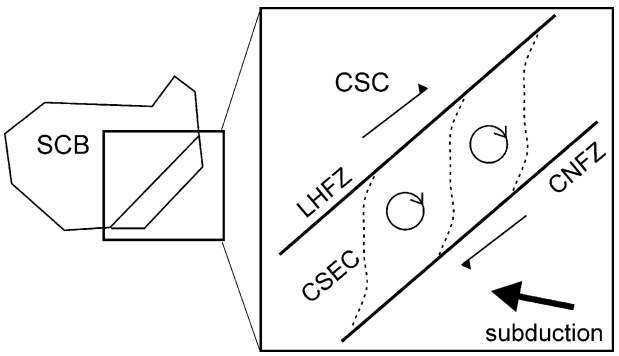


Fig. 12. The oblique subduction of the Pacific Plate underneath South China exerted dextral shear along SE China. This dextral shear probably resulted in right-lateral strike-slip movement of blocks along two marginal–parallel faults in SE China. The accumulated, penetrative shear may eventually separate the CSEC sliver into several microblocks, which are located between the LHFZ and the CNFZ, and rotate these microblocks in a clockwise sense about vertical-axes.

margin of South China may result in not only dextral movements along the two marginal parallel faults, but also the separation of the CSEC into several micro-blocks that are subsequently rotated about vertical-axes in a clockwise sense (Fig. 12).

Alternatively, the clockwise rotation of the CSEC relative to the CSC inferred from paleomagnetic data could also be interpreted as indicating pure NW–SE extension along the southeastern margin of South China. Extension has been generally believed to be the dominant tectonic regime in SE China since the Late Cretaceous (Charvet et al., 1994; Lapierre et al., 1997; Li and McChulloch, 1998), as evidenced by the prominent NE-trending graben systems and a number of marginal basins in SE China (Gilder et al., 1991; Ren et al., 2002). The NW–SE extension attenuated the crust, stretched the CSEC, and probably displaced the CSEC as a whole southeastward since the Late Cretaceous. This continued

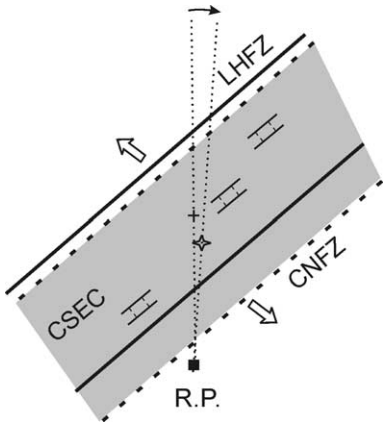


Fig. 13. The NW–SE marginal pure extension may have displaced the CSEC as a whole southeastward, generating an apparent clockwise rotation in paleomagnetic records. In this case, no real block rotations was involved. The cross and the star denote the center of the CSEC before and after extension, respectively. The dash line bounded shaded region indicates the post-extension CSEC. R.P., pole of rotation.

southeastward extension alone can produce an apparent clockwise rotation of the CSEC in paleomagnetic records without requiring real block rotations in SE China (Fig. 13). Mechanisms responsible for the pervasive extension in SE China remain controversial. The widespread extension in SE China may be associated with the subduction of the proto-Pacific Plate. The extension probably started with a back–arc extension (Guo, 1983) and was sustained by continued subduction as the oceanic slab steepened (Pigott and Ru, 1994). The steepening dip of the subducting oceanic plate is also believed to have caused oceanward younging of the volcanic and granitic rocks in SE China (Zhou and Li, 2000). On the other hand, the extension may have resulted from the India–Asia collision, which extruded the Indochina Block toward the southeast and rotated it in a counterclockwise sense, generating transtension in marginal SE China (Tapponnier et al., 1982, 1986).

5. Conclusions

Investigation of Upper Cretaceous dikes in Hong Kong has provided new information for the Cretaceous paleomagnetic database for Coastal SE China. The new information has forced a reassessment of the paleopole dataset for granitic intrusions in Hong Kong, which were previously believed to carry the Late Cretaceous remanence (Chan, 1991). The recently revised stratigraphy of Hong Kong suggests that these granitic intrusions are 25–30 million years older than previously thought and they carry Late Jurassic to Early Cretaceous remanences. Directional discrepancy between the paleomagnetic data of granitic rocks and dikes indicates a pre-Cretaceous counter-clockwise rotation, which occurred, at least, locally in Hong Kong if not in all SE China.

A synthesis of the Cretaceous paleopoles from the South China Block found discordant Cretaceous paleopoles between the CSEC and the CSC. The apparent separation of Cretaceous poles from the SCB implies a $11.6 \pm 9.3^{\circ}$ clockwise rotation of the CSEC relative to the stable CSC. This post-Cretaceous clockwise rotation was probably widespread in southeastern China and has been interpreted as indicating the following kinematics:

1. Vertical-axis clockwise rotation of microblocks located between the LHFZ and the CNFZ since the Late Cretaceous because of the associated dextral shear exerted along the South China margin during the subduction of the Pacific Plate underneath the South China Block; OR
2. Regional NW–SE extension of the South China margin probably displaced the CSEC as a whole toward the southeast, generating an apparent clockwise rotation in paleomagnetic records with no physical block rotations.

Although the positive fold test and the presence of dual polarities argue that the dikes carry a Late Cretaceous remanence, it is important to note that the tilt variation around the fold may not be great enough for a fully definitive fold test. In addition, only four cooling units were investigated, which may not guarantee an adequate averaging of secular variation. Therefore, it would be beneficial to acquire more paleomagnetic data from cooling units in other parts of the CSEC in future studies. It would also be interesting to test whether the pre-Cretaceous counter-clockwise rotation is a regional feature of the CSEC when more Jurassic poles are obtained from this region. Despite the limitations of the present study, the sparse paleomagnetic data currently available from the CSEC make the paleomagnetic results of this study an important contribution to the understanding of the tectonic problems in this region.

Acknowledgements

This work was supported by Hong Kong RGC award to LSC. We are grateful to Dr Randolph J. Enkin and an anonymous reviewer for their helpful comments that improved the manuscript. We would also like to thank Kenneth P. Kodama for helpful review prior to submission. Gavin Chan and Walter Tsang provided some assistance in the field. Liangshu Wang and Shaowen Liu helped collect some Chinese references.

References

Allen, P.M., Stephens, E.A., 1971. Report on the geological survey of Hong Kong. Hong Kong Government Press, Hong Kong, pp. 116 plus 2 maps.

Besse, J., Courtillot, V., 1991. Revised and synthetic apparent polar wnader paths of the African Eurasian, North American and Indian Plates, and true polar wander since 200 Ma. Journal of Geophysical Research 96, 4029–4050.

Burchfiel, B.C., Chen, Z., Liu, Y., Royden, L.H., 1995. Tectonics of the Long.men Shan and adjacent regions, central China. International Geology Review 37, 661–735.

Campbell, S.D.G., Sewell, R.J., 1998. A proposed revision of the volcanic stratigraphy and related plutonic classfication of Hong Kong. Hong Kong Geologist 4, 1–11.

Chan, L.S., 1991. Paleomagnetism of late Mesozoic granite intrusions in Hong Kong: implications for Upper Cretaceous reference pole of South China. Journal of Geophysical Research 96, 327–335.

Charvet, J., Lapierre, H., Yu, Y., 1994. Geodynamic significance of the Mesozoic volcanism of southeastern China. Journal of Southeast Asia Earth Sciences 9 (4), 387–396.

Chen, A., 1999. Mirror-image thrusting in the South China Orogenic Belt: tectonic evidence from western Fujian, southwestern China. Tectonophysics 305, 497–519.

Chen, Z.H., Li, J.Y., Sewell, R.J., 1999. The basaltic dikes in coastal Fujian, Hong Kong and northern Taiwan, Abstract In: Symposium of joint conference in geosciences of China, Hong Kong, and Taiwan, August, pp. 30.

Cheng, G.L., Sun, Y.H., Sun, Q.G., Wang, L.H., 1996. Phanerozoic paleomagnetic data for the main blocks of China. Seismology and Geology 18 (2), 182–193.(in Chinese with English abstract).

Cisowski, S.M., Dunn, J.R., Fuller, M., Wasilewski, P.J., 1990. NRM:IRM(s) demagnetization plots of intrusive rocks and the origin of their NRM. Tectonophysics 184, 35–54.

Davis, D.W., Sewell, R.J., Campbell, S.D.G., 1997. U–Pb dating of Mesozoic igneous rocks from Hong Kong. Journal of the Geological Society, London 154, 1067–1076.

Enkin, J.R., Courtillot, V., Xing, L., Zhang, Z., Zhuang, Z., Zhang, J., 1991. The stationary Cretaceous paleomagnetic pole of Sichuan (South China Block). Tectonics 10, 547–559.

Enkin, R.J., Yang, Z.Y., Courtillot, V., 1992. Paleomagnetic constraints on the geodyanmic history of the major blocks of China from the Permian to the present. Journal of Geophysical Research 97 (13), 953–989.

Fisher, R.A., 1953. Dispersion on a sphere. Proceedings of the Royal Society of London A 217, 295–305.

Fuller, M., Cisowski Hart, M., Haston, R., Schnidtke, E., Jarrard, R., 1988. NRM:IRM(s) demagnetization plots; an aid to the interpretation of natural remanent magnetization. Geophysical Research 184, 35–54.

Funahara, S., Nishiwaki, N., Miki, M., Murata, F., Otofuji, Y., Wang, Y.Z., 1992. Paleomagnetic study of Cretaceous rocks from the Yangtze block, central Yunna, China: implications for the India–Asia collision. Earth and Planetary Science Letters 113, 77–91.

Gilder, S.A., Keller, R.G., Luo, M., Goodell, P.C., 1991. Timing and spatial distribution of rifting in China. Tectonophysics 197, 225–243.

Gilder, S.A., Coe, R.S., Wu, H.R., Kuang, G.D., Zhao, X.X., Wu, Q., Tang, X.Z., 1993. Cretaceous and Tertiary paleomagnetic results from southeast China and their tectonic implications. Earth and Planetary Science Letters 117, 637–652.

Gilder, S.A., Leloup, P.H., Courtillot, V., Chen, Y., Coe, R., Zhao, X.X., Xiao, W.J., Halim, N., Cogné, J.P., Zhu, R., 1999. Tectonic evolution of Tancheng-Lujiang (Tan-Lu) Fault via Middle Triassic to Early Cenozoic paleomagnetic data. Journal of Geophysical Research 15, 365–390.

Gu, X.L., and Gu, W., 1990. Paleomagnetism and tectonic implication of the Late Cretaceous Red' Bed in the Ji-tai Basin, South China, Abstracts for the 4th National Symposium on Paleomagnetism, p22, Zhejiang University, Hangzhou, China, in Chinese.

Guo, L.Z., Shi, Y.S., Ma, R.S., 1983. Formations and evolution of Meso-Cenozoic active continental margin and island arcs of west pacific. Acta Geologica 57 (1), 11–18.in Chinese.

Hu, L., Li, P., Ma, X., 1990. A magnetostratigraphic study of Cretaceous Red Beds from Shanghan, western Fujian, China, Geology of Fujian 1, 33–42. in Chinese.

Huang, K., Opdyke, N.D., 1992. Paleomagnetism of Cretaceous to lower Tertiary rocks from southwestern Sichuan: a revisit. Earth and Planetary Science Letters 112, 29–40.

Jahn, B.M., Zhou, X.H., Li, J.L., 1990. Formation and tectonic evolution of Southeastern China and Taiwan: Isotopic and geochemical constraints. Tectonophysics 183, 145–160.

Kent, D.V., Xu, G., Huang, K., Zhang, W.Y., Opdyke, N.D., 1986. Paleomagnetism of Upper Cretaceous rocks from South China, Earth and Planetary Science Letters 79, 179–184.

Kirschvink, J.L., 1980. The least-square line and plane and the analysis of paleomagnetic data. Geophysical Journal of Royal Astronomy Society 62, 699–718.

Lapierre, H., Jahn, B.M., Charvet, J., Yu, Y.W., 1997. Mesozoic felsic arc magmatism and continental olivineTholeiites in Zhejiang Province and their relationship with the tectonic activity in southeastern China. Tectonophysics 274, 321–338.

Li, Z.X., 1998. Tectonic history of the major East Asia lithospheric blocks since the Mid-Proterozoic—A synthesis. In: Flower, M., Chung, S.L., Lee, T.Y., Lo, C.H. (Eds.), Mantle Dynamics and Plate Interactions in East Asia, AGU Geodynamics Series No 27, 27., pp. 221–243.

Li, X.H., and McCulloch, M.T., 1998. Geochemical characteristics of Cretaceous mafic dikes from northern Guangdong, SE China: age, origin and tectonic significance. Mantle dynamics and plate interactions in East Asia. Geodynamics 27, 405–419.

Lin, J., and Xie, M., 1996. Time and space distributions of seismicity in southeastern coastal China. Seismicity in East Asia, Geological Society of Hong Kong, Bulletin Number 5, 12–22.

Lin, J.Z., Wei, B.L., Xue, J.M., 1984. The source process of the 1981 earthquake swarm, Haifeng, Guangdong. In: A Collection of Papers from the International Symposium on Continental Seismicity and Earthquake Prediction (ISCEP), Seismological Press, Beijing, pp. 89–99.

Liu, Y.Y., Morinaga, H., 1999. Cretaceous paleomagnetic results from Hainan Island in south China supporting the extrusion model of Southeast Asia. Tectonophysics 301, 133–144.

Liu, Y.Y., Tian, W.H., Zhu, Y.M., 1999. Paleomagnetic study of Cretaceous basin I at Lishui, Zhejiang Province. Earth Science-Journal of China University of Geosciences 24 (2), 139–141.in Chinese with English abstract.

Lu, H.F., Jia, D., Wang, Z.H., Guo, L.Z., Shi, Y.S., 1993. The composition and collision history of the Min-Tai microcontinent. In: Li, J.L., (Ed.), Lithospheric Structure and Evolution of Southeastern Continent, 12-26. Metallurgical industry press, Beijing, pp. 12–26.

Ma, X., Yang, Z., Xing, L., 1993. The lower Cretaceous reference pole for North China, and its tectonic implications. Geophysical Journal International 115, 323–331.

McCarthy, A.J., Elders, C.F., 1997. Cenozoic deformation in Sumatra: oblique subduction and the development of the Sumatran Fault system. In: Fraser, A.J., (Ed.), Petroleum Geology of Southeast Asia. Journal of Geological Society, London, Special Publication 126, pp. 355–363.

Mcfadden, P.L., Jones, D.L., 1981. The fold test in paleomagnetism. Geophysical Journal of Royal Astronomy Society 67, 53–58.

Metcalfe, I., 1996. Pre-Cretaceous evolution of SE Asian terranes. In: Hall, R., Blundell, D. (Eds.), Tectonic evolution of Southeast Asia, vol. 106., pp. 97–122.

Otofuji, Y., Liu, Y., Yokoyama, M., Tamai, M., Yin, J., 1998. Tectonic deformation of the southwestern part of the Yangtze craton inferred from paleomagnetism. Earth and Planetary Science Letters 156, 47–60.

Pigott, J.D., Ru, K., 1994. Basin superposition on the northern margin of the South China Sea. Tectonophysics 235, 27–50.

Ren, J., Tamaki, K., Li, S., Zhang, J., 2002. Late Mesozoic and Cenozoic rifting and its dynamic setting in Eastern China and adjacent areas. Tectonophysics 344, 175–205.

Sato, K., Liu, Y., Zhu, Z., Yang, Z., Otofuji, Y., 1999. Paleomagnetic study of middle Cretaceous rocks from Yunlong, western Yunnan, China: evidence of southwards displacement of Indochina. Earth and Planetary Science Letters 165, 1–15.

Sewell, R.J., Campbell, S.D.G., 1997. Geochemistry of coeval Mesozoic plutonic and volcanic suites in Hong Kong. Journal of the Geological Society, London 154, 1053–1066.

Sewell, R.J., Campbell, S.D.G., Davis, D.W., 1998. Three new U–Pb ages from igneous rocks in the NE New Territories of Hong Kong and their structural significance. Hong Kong Geologist 4, 31–36.

Sewell, R.J., Campbell, S.D.G., Fletcher, C.J.N., Lai, K.W., Lai, P.A., 2000. The pre-Quaternary geology of Hong Kong, Geotechnical Eng. Off. Civil Eng. Dept., Government of the Hong Kong Spec. Admin. Region, pp. 181.

Tapponnier, P., Peltzer, G., Le Dain, A.Y., Armijo, R., Cobbold, P., 1982. Propagating extrusion tectonics in Asia: New insights from simple experiments with plasticine. Geology 10, 611–616.

Tapponnier, P., Peltzer, G., Armijo, R., 1986. On the mechanics of the collision between India and Asia. In: Coward, M.P., Ries, A.C. (Eds.), Collision tectonics, Geological Society of London, Special Publication 19, pp. 115–157.

Van der Voo, R., 1990. The reliability of paleomagnetic data. Tectonophysics 184, 1–9.

Wang, Z.H., Lu, H.F., 1997. Evidence and dynamics for the change of strike-slip direction of the Changle Nanao ductile shear zone, southeastern China. Journal of Asian Earth Sciences 15, 507–515.

Weng, W.H., 1927. Earth's crust movement and volcanism in eastern China since Mesozoic. Acta Geologica Sinica 6, 9–36.

Yang, Z., Yin, J., Sun, Z., Otofuji, Y., Sato, K., 2001. Discrepant Cretaceous paleomagnetic pole between Eastern China and Indochina: a consequence of the extrusion of Indochina. Tectonophysics 334, 101–113.

Yu, Z.Z., Yang, S.F., Guo, L.Z., Shi, Y.S., 1990. Paleomagnetic results from the Hainan Terrane and their geological implications, South China. Abstracts for the 4th National Symposium on Paleomagnetism, pp. 23–25, Zhejiang University, Hangzhou, China. in Chinese.

Zhai, Y.J., Zhou, X.X., 1989. Paleomagnetism and tectonic evolutions of North and South China blocks since the Phanerozoic. Acta Geophysica Sinica 32 (3), 292–307.in Chinese with English abstract.

Zhai, Y.Z., Seguin, M.K., Zhou, Y.X., Dong, J.M., Zheng, Y.X., 1992. New paleomagnetic data from the Huanan block, China and Cretaceous tectonics in eastern China. Physics of Earth and Planetary Interior 73, 163–188.

Zhang, B.Z., 1993. Plate tectonics. Jilin Science and Technology Press, in Chinese.

Zhao, X.X., Coe, R.S., Gilder, S.A., Frost, G.M., 1996. Paleomagnetic constraints on the palaeogeography of China: implications for Gondwanaland. Australian Journal of Earth Sciences 43, 643–672.

Zhou, X.M., and Li, W.X., 2000. Origin of Late Mesozoic igneous rocks in Southeastern China: implications for lithosphere subduction and underplating of mafic magmas. Tectonophysics 326, 269–287.

Zhou, Z.X., Lu, L.Z., Zhang, B.M., 1988. Preliminary paleomagnetic results from Panzhihua-Xichang region. In: Zhang, Y.X. and Yuan, X.C. (Eds), A collection of papers regarding the rifting in Pan-Xi area, China, Vol. 3, pp. 212–230, Geological Publishing House, Beijing. in Chinese.

Zhu, Z., Hao, T., Zhao, H., 1988. Paleomagnetic study on the tectonic evolution of Pan-Xi block and adjacent area during Yinzhi-Yanshan period. Acta Geophysica Sinica 31, 420–431.

Zhuang, Z., Tian, D., Ma, X., Ren, X., Jiang, X., Xu, S., 1988, A., 1988. A paleomagnetic study along the Yuan-Tainquan Cretaceous-Eogene section in the Sichuan basin. Geophysical and Geochemical Exploration 12, 224–228. in Chinese.

Zijderveld, J.D.A., 1967. AC demagnetization of rocks: analysis of results. In: Collinson, D.W., Creer, K.M., Runcorn, S.K. (Eds.), Methods and Paleomagnetism, Elsevier, Amsterdam, pp. 245–268.